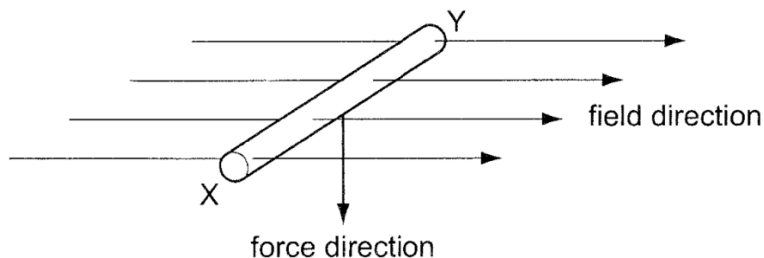


- 24 A current-carrying conductor is placed at right angles to a uniform magnetic field of flux density 0.50 T. A 10 cm length of conductor lies within the field and experiences a force of 2.4 mN.



What is the direction of electron flow and rate of flow of electrons in the conductor?

	direction of electron flow	rate of flow of electrons / s
A	X to Y	4.8×10^{-2}
B	Y to X	4.8×10^{-2}
C	X to Y	3.0×10^{17}
D	Y to X	3.0×10^{17}

Ans: D

Using FLHR, current is from X to Y. Direction of current flow is opposite to direction of electron flow.

$$F_B = BIL$$

$$0.0024 = (0.50)(I)(0.10)$$

$$I = 0.048$$

$$\frac{N}{t}e = 0.048$$

$$\begin{aligned}\frac{N}{t} &= \frac{0.048}{1.60 \times 10^{-19}} \\ &= 3.0 \times 10^{17}\end{aligned}$$